

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/10/2026	DATE SUBMITTED:	09/10/2026
NAME:	Christopher T. Fegalan	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 7.1 MANAGING FILES AND CREATING ROLES IN ANSIBLE

1. PROCEDURE OUTPUT

Server 1 (Ubuntu)	192.168.0.102
Server 2 (Ubuntu)	192.168.0.104
Server 3 (CentOS)	192.168.0.103

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it "files." Create a file inside that directory and name it "default_site.html." Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.

```
GNU nano 1.2 default_site.html
!DOCTYPE html>
html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
  </head>
  <body>
    <h1>Welcome to My Web Server</h1>
    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>
  </body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:

```
- name: copy default html file for site
  tags: apache, apache2, httpd
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

```
TASK [Mariadb - Restarting/Enabling CentOS] *****
skipping: [server2]

TASK [Install mariadb package (Ubuntu)] *****
ok: [server2]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [server3]

TASK [Install Samba Package] *****
ok: [server3]

PLAY RECAP *****
server1      : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescue=0
d=0 ignored=0
server2      : ok=3    changed=0    unreachable=0    failed=0    skipped=3    rescue=0
d=0 ignored=0
server3      : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescue=0
d=0 ignored=0

user123@workstation:~/CPE212_fegalan/H0A7$
```

```
skipping: [server2]

TASK [Install mariadb package (Ubuntu)] *****
ok: [server2]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [server3]

TASK [Install Samba Package] *****
ok: [server3]

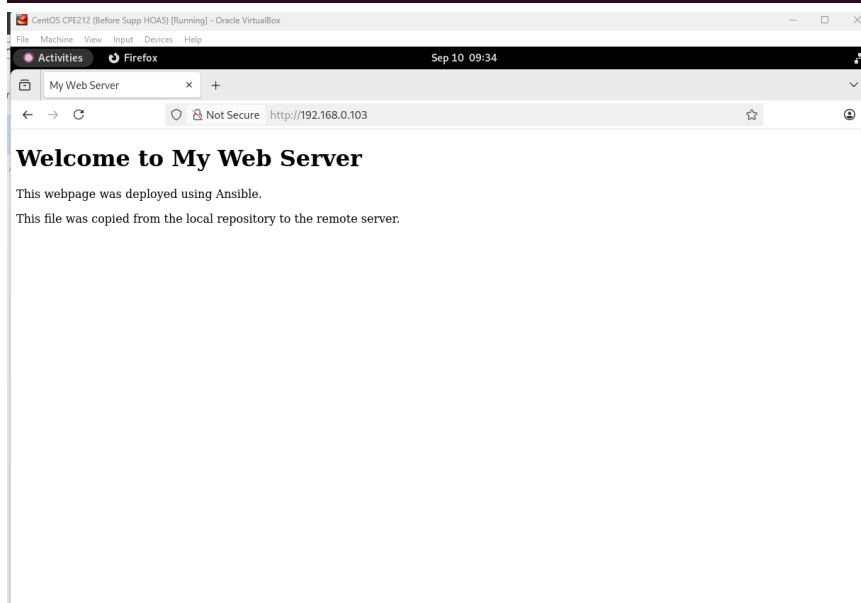
PLAY RECAP *****
server1      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescue=0
d=0 ignored=0
server2      : ok=3    changed=0    unreachable=0    failed=0    skipped=3    rescue=0
d=0 ignored=0
server3      : ok=7    changed=1    unreachable=0    failed=0    skipped=1    rescue=0
d=0 ignored=0
```

- I added server3 (CentOS) to the web_servers.

```
all:
  children:
    web_servers:
      hosts:
        server1:
          ansible_host: 192.168.0.102
        server3:
          ansible_host: 192.168.0.103
          ansible_user: centos_chris
    db_servers:
      hosts:
        server2:
          ansible_host: 192.168.0.104
    file_servers:
      hosts:
        server3:
          ansible_host: 192.168.0.103
          ansible_user: centos_chris
```

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

```
user123@server1:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale
=1.0">
    <title>My Web Server</title>
  </head>
  <body>
    <h1>Welcome to My Web Server</h1>
    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote
server.</p>
  </body>
</html>
```



5. Sync your local repository with GitHub and describe the changes.


CPE212_fegalan
Public





home
had recent pushes 29 seconds ago
Compare & pull request

home
2 Branches
0 Tags

Add file
Code

This branch is 3 commits ahead of `main`.
Contribute


christopherfegalan
Added: TASK1 HOA7
19646e6 · 1 minute ago
26 Commits

HOA4	HOA4,5,6 FIX	2 days ago
HOA5	HOA4,5,6 FIX	2 days ago
HOA6	Removed: install_apache.yaml	2 days ago
HOA7	Added: TASK1 HOA7	5 minutes ago
.notes.txt.swp	remove	2 days ago
README.md	Updated README.md	2 months ago
notes.txt	update: added notes.txt	2 days ago

CPE212_fegalan / HOA7




christopherfegalan
Added: TASK1 HOA7
34b31e · 13 minutes ago
History

This branch is 3 commits ahead of `main`.
Contribute

Name	Last commit message	Last commit date
...		
files	Added: TASK1 HOA7	13 minutes ago
homeinventory.yaml	Added: TASK1 HOA7	13 minutes ago
site.yaml	Added: TASK1 HOA7	13 minutes ago

- HOA 7 directory was added to my repository along with the necessary files for this activity.

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play:


user123@workstation: ~/CPE212_fegalan/HOA7





GNU nano 7.2
site.yaml

```

- hosts: all
  become: true

- hosts: workstations
  become: true

tasks:
  - name: install unzip
    package:
      name: unzip

  - name: install terraform
    unarchive:
      src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
      dest: /usr/local/bin
      remote_src: yes
      mode: 0755
      owner: root
      group: root

```

2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address.

```
GNU nano 7.2 homeinventory.yaml
server1:
  ansible_host: 192.168.0.102
server3:
  ansible_host: 192.168.0.103
  ansible_user: centos_chris
db_servers:
  hosts:
    server2:
      ansible_host: 192.168.0.104
file_servers:
  hosts:
    server3:
      ansible_host: 192.168.0.103
      ansible_user: centos_chris
workstations:
  hosts:
    server1:
      ansible_host: 192.168.0.102
    server2:
      ansible_host: 192.168.0.104
```

3. Run the playbook. Describe the output.

```
TASK [Mariadb - Restarting/Enabling CentOS] *****
skipping: [server2]

TASK [Install mariadb package (Ubuntu)] *****
ok: [server2]

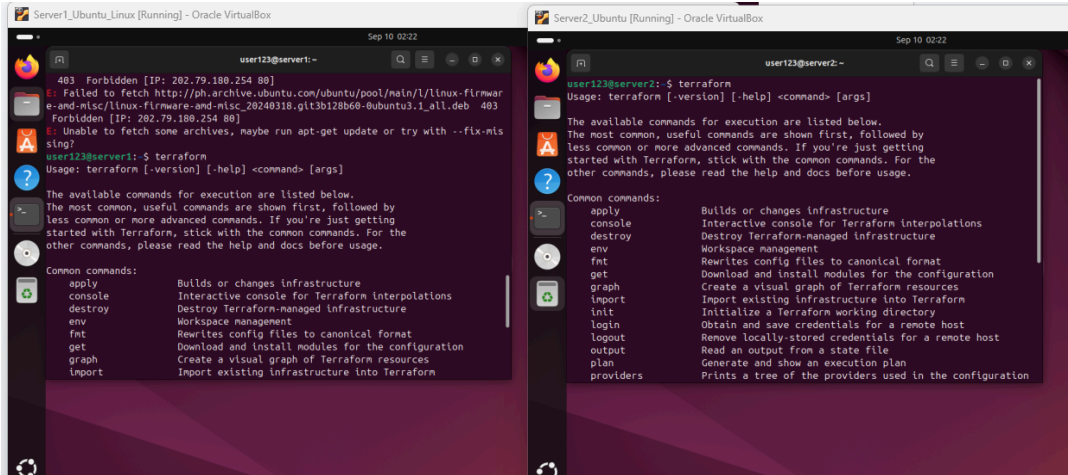
PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [server3]

TASK [Install Samba Package] *****
ok: [server3]

PLAY RECAP *****
server1      : ok=7    changed=1    unreachable=0    failed=0    skipped=2    resc
d=0    ignored=0
server2      : ok=6    changed=1    unreachable=0    failed=0    skipped=3    resc
d=0    ignored=0
server3      : ok=7    changed=0    unreachable=0    failed=0    skipped=1    resc
d=0    ignored=0
```

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.



- For both my ubuntu servers it displayed the terraform's standard help manual.

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)

```
GNU nano 7.2 site2.yml
---
- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    - db_servers
```

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

```
user123@workstation:~/CPE212_fegalan/H0A7$ mkdir roles
user123@workstation:~/CPE212_fegalan/H0A7$ cd roles
user123@workstation:~/CPE212_fegalan/H0A7/roles$ mkdir base, web_servers, file_servers, db_servers
, workstations
user123@workstation:~/CPE212_fegalan/H0A7/roles$ ls
base, db_servers, file_servers, web_servers, workstations
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

```
GNU nano 7.2 main.yml *
---
- name: Install Updates (CentOS)
  tags: always
  dnf:
    update_only: yes
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: Install Updates (Ubuntu)
  tags: always
  apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
```

- base main.yml

```
GNU nano 7.2 db_servers/tasks/main.yaml *
---
- name: Install mariadb package (CentOS)
  tags: centos,db,mariadb
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: Reload system daemon
  systemd:
    daemon_reload: yes
  when: ansible_distribution == "CentOS"

- name: "Mariadb - Restarting/Enabling CentOS"
  service:
    name: mariadb
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"

- name: Install mariadb package (Ubuntu)
  tags: db,mariadb,ubuntu
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"
```

```
GNU nano 7.2 file_servers/tasks/main.yaml
---
- name: Install Samba Package
  tags: samba
  package:
    name: samba
    state: latest
```

```
GNU nano 7.2 web_servers/tasks/main.yaml
---
- name: install apache and php for Ubuntu Servers
  tags: apache,apache2,ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS Servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache,centos,httpd
  service:
    name: httpd
    state: started
    enabled: yes
  when: ansible_distribution == "CentOS"

- name: copy default html file for site
  tags: apache, apache2, httpd
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

```
GNU nano 7.2                                workstations/tasks/main.yaml
---
- name: install unzip
  package:
    name: unzip

- name: install terraform
  unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
    dest: /usr/local/bin
    remote_src: yes
    mode: 0755
    owner: root
```

4. Run the site.yml playbook and describe the output.

```
user123@workstation: ~/CPE212_fegalan/HOA7

TASK [web_servers : install apache and php for CentOS Servers] *****
skipping: [server1]
ok: [server3]

TASK [web_servers : start httpd (CentOS)] *****
skipping: [server1]
ok: [server3]

TASK [web_servers : copy default html file for site] *****
ok: [server1]
ok: [server3]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [server2]

TASK [db_servers : Install mariadb package (CentOS)] *****
skipping: [server2]

TASK [db_servers : Reload system daemon] *****
skipping: [server2]

TASK [db_servers : Mariadb - Restarting/Enabling CentOS] *****
skipping: [server2]

TASK [db_servers : Install mariadb package (Ubuntu)] *****
ok: [server2]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [server3]

TASK [file_servers : Install Samba Package] *****
ok: [server3]

PLAY RECAP *****
server1      : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2      : ok=5    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server3      : ok=6    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- For this, I took out pre tasks/base role since I keep getting 404 error.
- Also from what i can observe after running the site2.yaml (copy of site.yaml that has roles). By calling the hosts along with the roles in the site2.yaml playbook it automatically understood to look for the task directory and ran the main.yaml file inside it. Simplifying the readability of the playbook file and segregating tasks based on which role needs it.

GITHUB:


CPE212_fegalan Public

Pin

main
2 Branches
0 Tags

Add file
Code


christopherfegalan FIN: HOA7

6ca74c9 · 2 minutes ago
27 Commits

HOA4	HOA4,5,6 FIX	2 days ago
HOA5	HOA4,5,6 FIX	2 days ago
HOA6	Removed: install_apache.yaml	2 days ago
HOA7	FIN: HOA7	2 minutes ago
.notes.txt.swp	remove	2 days ago
README.md	Updated README.md	2 months ago
notes.txt	update: added notes.txt	2 days ago

CPE212_fegalan / HOA7

Add file
...


christopherfegalan FIN: HOA7

6ca74c9 · 2 minutes ago
History

Name	Last commit message	Last commit date
..		
files	Added: TASK1 HOA7	1 hour ago
roles	FIN: HOA7	2 minutes ago
homeinventory.yaml	FIN: HOA7	2 minutes ago
site.yaml	FIN: HOA7	2 minutes ago
site2.yaml	FIN: HOA7	2 minutes ago

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?

- Creating roles breaks complex playbooks into small, reusable components, making the code clean, organized, and easy to maintain as the main system/infrastructure grows.

2. What is the importance of managing files?

- Managing files makes sure that all servers receive identical configurations and binaries while being able to enforce strict security through proper ownership and access permissions.

Conclusions/Learnings:

- This activity helped me better understand why ansible is so useful for managing servers. Going through this activity and trying to manage files through copying/unarchiving setting their owner, group, mode(permission) that the remote server would have after being copied or downloaded to the remote machines were very interesting to do. Creating roles and learning the structure for it, needing a group directory or the role and inside it needing a task directory and inside that needing the tasks or the main.yaml to execute. Which was very helpful to modularize tasks and don't have a very convoluted playbook unlike last activity having tons of tasks in site.yaml that I sometimes get confused which part I am reading.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.